Run report: genus-4 screen for étale covers at K-283, K-409 and K-571 (lead L8)

run-20261018T2144Z · study `research/ecdlp_leads_20261018` · commit `4369a50a`

Kind. This is an exact enumeration with a screen on a necessary condition, run with the native Rust crate `etale_search` (`screen.rs`, `genus4.rs`). It is not a timing or statistical measurement. Host: shared cloud container (Xeon 2.80 GHz, 4 cores, rustc 1.97.0), isolation L0. It builds on run-20261018T2129Z, which closed the étale route for bases of genus ≤ 3.

Why genus 4. The L(1/2) budget (derived) allows base genus $g_0 \leq 4$ for K-283, ≤ 5 for K-409 and ≤ 7 for K-571, so genus 4 decides K-283. An $\mathbb{F}_2$-rational, geometrically $\mathbb{Z}/n$-Galois cover with Frobenius acting by λ exists iff $n \mid P_B(\lambda)$, and T_n in the Prym needs ord $\lambda \leq g_0 - 1 = 3$. The run tests λ of order 1, 2 and 3.

What ran.

- **Hyperelliptic models:** $y^2 + h(x)y = f(x)$ with $\deg h \leq 5$ and $\deg f \leq 10$.
- **Canonical models:** $X = Q \cap C$ in $\mathbb{P}^3$, with Q one of three normal forms (hyperbolic, elliptic, cone; Xarles, Lemma 9), and the cubic C taken modulo $Q \cdot L$ over the 16 monomials not divisible by $x_0 x_1$.
- **Smoothness:** tested over $\mathbb{P}^3(\mathbb{F}_{16})$, $\mathbb{P}^3(\mathbb{F}_{32})$ and $\mathbb{P}^3(\mathbb{F}_{64})$. Every singular point has degree ≤ 6 (derived).
- **L-polynomials:** from the counts over $\mathbb{F}_2, \ldots, \mathbb{F}_{16}$, checked against the Weil bounds and, for canonical curves, against out-of-sample counts N_5 and N_6.

quantity	hyperelliptic	canonical (trigonal)	all genus 4
smooth models	49,152	55,368	104,520
distinct L-polynomials	217	521	620 (118 shared)
largest $\#J = L(1)$	139	260	**260**
largest $\lvert L(-1)\rvert$	139	144	**144**
largest $\#X(\mathbb{F}_2)$	6	8	8
N_5, N_6 checked / failed	–	110,736 / 0	
$n \mid L(\pm 1), n \in \{283, 409, 571\}$	none	none	**none**
$n \mid P(\lambda)$, ord $\lambda = 3$	283, 409	283, 409	

Census check (sourced). Xarles (arXiv:2007.07822, Theorem 10 and Tables 1 and 5) gives 620 L-polynomials for genus-4 curves over $\mathbb{F}_2$. Of these, 403 do not occur for hyperelliptic curves and 99 do not occur for trigonal curves. That leaves 217 hyperelliptic, 521 trigonal and 118 shared, all matched here. The largest numbers of points, 6 and 8, also match. The hyperelliptic model count is $384 \cdot 2^7$, the mass of $H_4(\mathbb{F}_2)$.

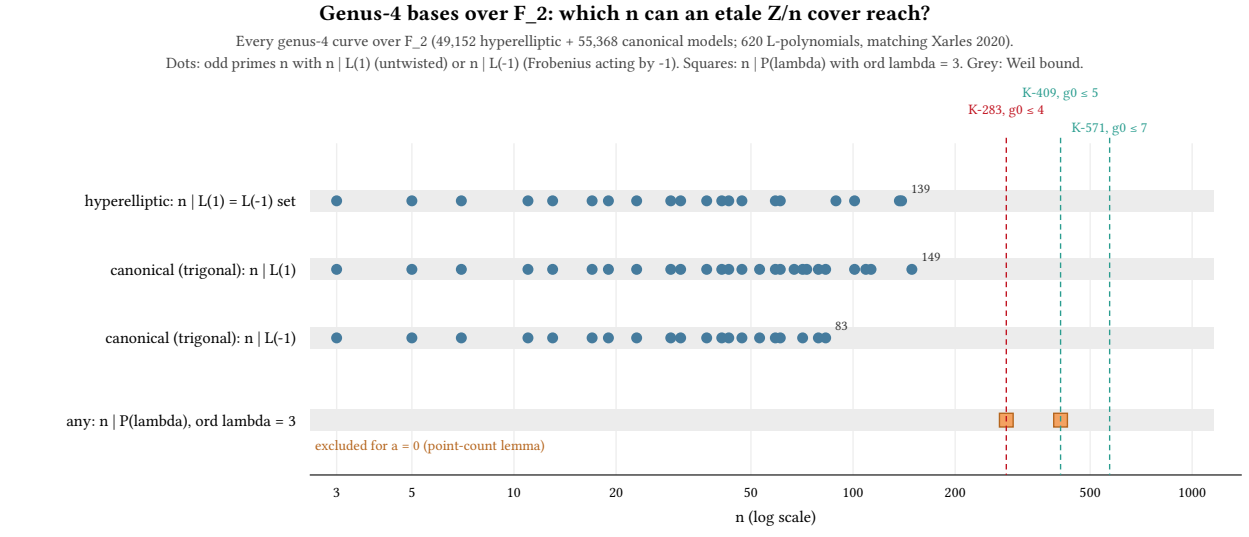


Figure 1: Odd primes dividing $L_B(\pm 1)$ over every genus-4 curve over $\mathbb{F}_2$, and the n with $n \mid P_B(\lambda)$ for λ of order 3, against the deployed sizes whose budget allows $g_0 = 4$. Recorded values from `reach.csv`.

Point-count lemma (derived). Suppose Frobenius acts by λ of order $k = g_0 - 1$ and T_n is in the Prym.

- The Prym's eigenvalues are closed under μ_k and contain those of T_n. A dimension count then shows they are exactly $\omega\zeta^j\pi^{\pm}$.
- So $\#D(\mathbb{F}_{2^k}) = \#B(\mathbb{F}_{2^k}) + kt_k$.
- The cover is untwisted over $\mathbb{F}_{2^k}$, so this number is $\equiv 0 \bmod n$.
- For n above the Hasse–Weil range this forces $\#B(\mathbb{F}_{2^k}) = -kt_k$.

For the $a = 0$ Koblitz curve, $t_3 = 5$, $t_4 = 1$ and $t_6 = 9$ are positive. So the maximal twist is impossible at $g_0 = 4, 5, 7$, and every order-3 candidate above is excluded.

What this closes. K-283 is closed: genus 2 and 3 fail the Weil bound, and genus 4 fails by measurement plus the lemma. Genus 4 is also closed for K-409 and K-571. These two remain open at $g_0 = 5$ ($k \le 3$), and K-571 also at $g_0 = 6, 7$. Together with the earlier runs, the étale route is closed for ECC2K-130, K-163, K-233 and K-283.

Not covered. Genus ≥ 5 bases, and ramified (L7) and non-cyclic covers.

Records. `RUN.txt` (commit, host, SHA-256), `genus4.jsonl`, `reach.csv` (via `tables.awk`), `reach.plot.awk`.